

To solve the problem, we analyze the conditions for a multiset S to have a specific index and determine the required parameters for the maximum possible index.

0.0.1 1. Condition for Index n

For a multiset S to have index at least n , it must be partitionable into multisets A_i and B_i such that $A_i = i + B_i$ for all $1 \leq i \leq n$ and for some $M < 20$. Since A_i and B_i partition S , $|S|$ must be even. This implies that for every $k \in \{1, \dots, n\}$, S must be decomposable into pairs of elements with difference k .

A necessary condition for such a partition is that the number of elements of S in each residue class modulo k must be even. This parity constraint is particularly restrictive as k increases.

0.0.2 2. Determining the Maximum Index N

If S contains 1, then 1 must be the smaller element in a pair for every $i \leq n$ (since $1 - i \leq 0$). Thus, $1 + i \in S$ for all $i \leq n$. This requires $M \geq n + 1$.

- For $n = 3$, S must contain $\{1, 2, 3, 4\}$. The multiset $S = \{1, 2, \dots, 12\}$ has index 3.
- For $n = 4$, S must satisfy $i = 4$ and $i = 3$ parity constraints. In the range $1 \leq x \leq 19$, it is impossible to satisfy the even-count requirements for moduli 2, 3, and 4 simultaneously while maintaining connectivity for $i = 1$ and $i = 2$ without exceeding the element limit of 19 or requiring significantly larger sets.
- For $n \geq 10$, a contradiction arises: 1 requires $1 + 9 = 10$ to be in S , but for $i = 10$, the element 10 would be isolated as $10 - 10 = 0$ and $10 + 10 = 20$ are both outside the allowed range. Thus, the maximum index is $N = 3$.

0.0.3 3. Minimizing M and Size $|S|$

For $N = 3$, we must find the smallest M (maximum element) and then the smallest size $|S|$.

- **Minimizing M :** To support index 3, S must support $i = 1, 2, 3$. The set $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ successfully supports these three partitions. Testing intervals of size $M < 12$ (such as $\{1, \dots, 11\}$) shows that they fail the parity or connectivity constraints for at least one $i \in \{1, 2, 3\}$. For example, $\{1, \dots, 11\}$ fails $i = 2$ partition matching and $i = 3$ parity in residue classes.
- **Minimizing Size $|S|$:** For $M = 12$, the smallest possible multiset that satisfies all conditions is $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$. Any removal of elements would either break the connectivity required for $i = 1$ or violate the parity requirements for residues modulo 2 or 3.

For index $N = 3$, the minimum maximum element M is 12, and the smallest size of such a multiset is 12.